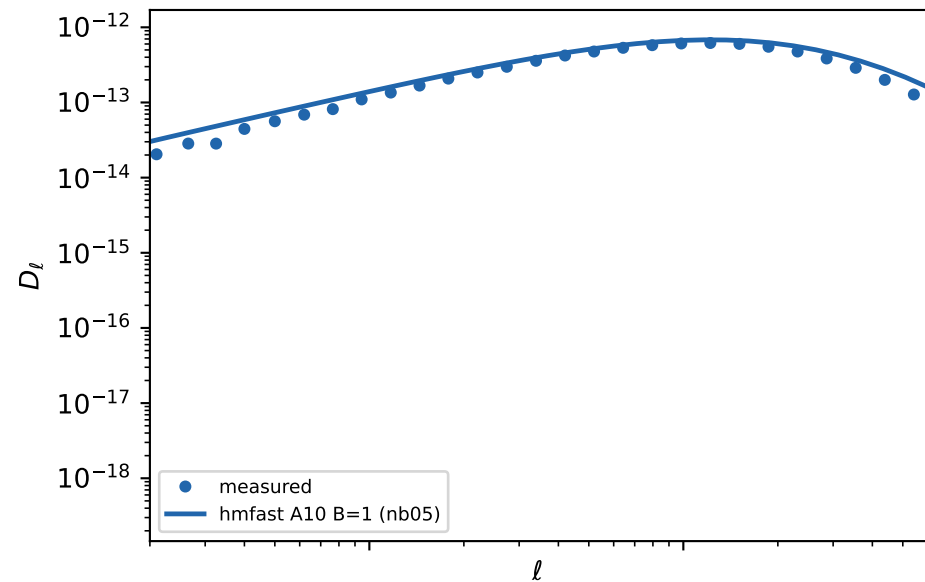
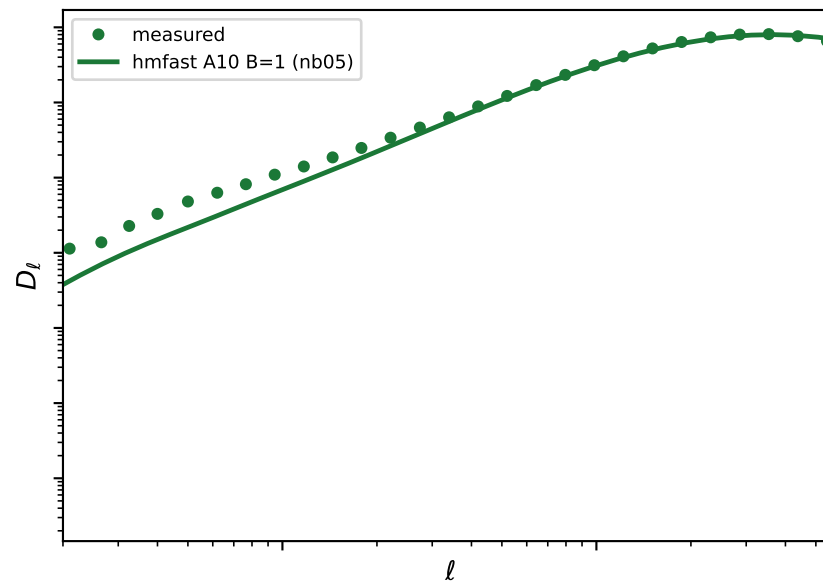
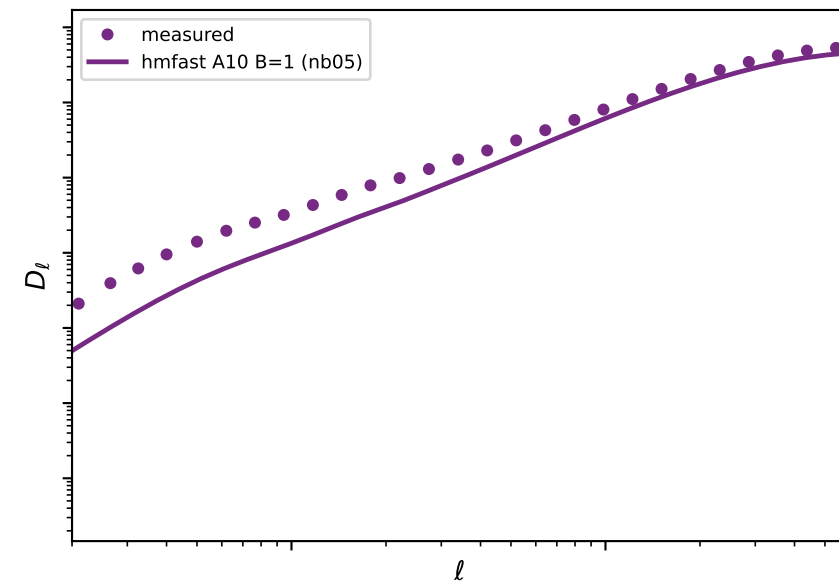
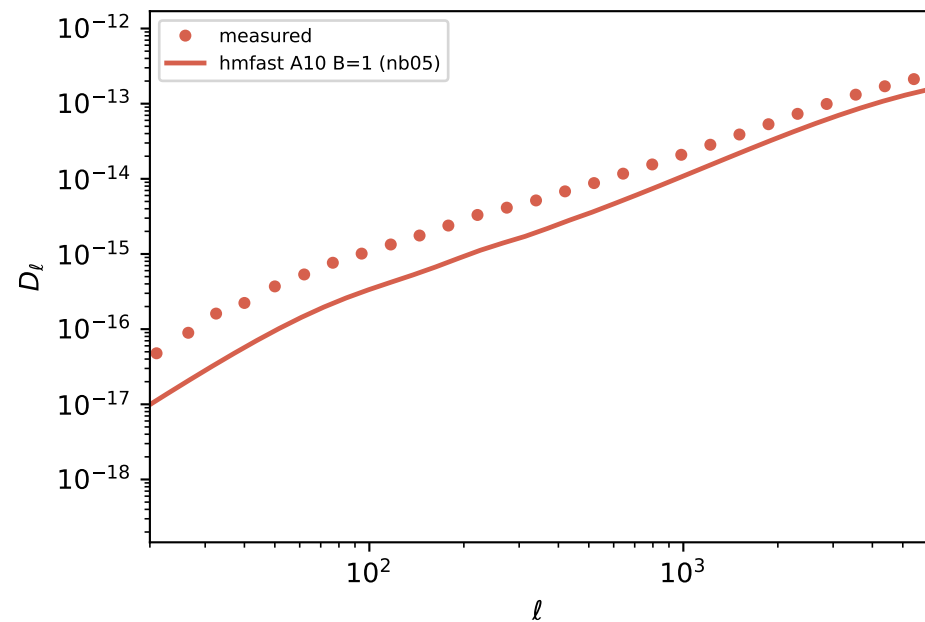


group 0: shells 0-6
 $z \in [0.00, 0.35]$, identitygroup 1: shells 7-15
 $z \in [0.35, 0.80]$, $\theta = 2.118$, $\phi = 0.964$ group 2: shells 16-26
 $z \in [0.80, 1.35]$, $\theta = 1.291$, $\phi = 1.748$ group 3: shells 27-43
 $z \in [1.35, 2.20]$, $\theta = 5.692$, $\phi = 0.563$ group 4: shells 44-59
 $z \in [2.20, 3.00]$, $\theta = 3.797$, $\phi = 1.455$ 